

CHAPTER C18

SEISMIC DESIGN REQUIREMENTS FOR STRUCTURES WITH DAMPING SYSTEMS

C18.1 GENERAL

The requirements of this chapter apply to all types of damping systems, including both displacement-dependent damping devices of hysteretic or friction systems and velocity-dependent damping devices of viscous or viscoelastic systems (Soong and Dargush 1997, Constantinou et al. 1998, Hanson and Soong 2001). Compliance with these requirements is intended to produce performance comparable to that for a structure with a conventional seismic force-resisting system, but the same methods can be used to achieve higher performance.

The damping system (DS) is defined separately from the seismic force-resisting system (SFRS), although the two systems may have common elements. As illustrated in Fig. C18.1-1, the DS may be external or internal to the structure and may have no shared elements, some shared elements, or all elements in common with the SFRS. Elements common to the DS and the SFRS must be designed for a combination of the loads of the two systems. When the DS and SFRS have no common elements, the damper forces must be collected and transferred to members of the SFRS.

C18.2 GENERAL DESIGN REQUIREMENTS

C18.2.1 System Requirements. Structures with a DS must have an SFRS that provides a complete load path. The SFRS must comply with all of the height, Seismic Design Category, and redundancy limitations and with the detailing requirements specified in this standard for the specific SFRS. The SFRS without the damping system (as if damping devices were disconnected) must be designed to have not less than 75% of the strength required for structures without a DS that have that type of SFRS (and not less than 100% if the structure is horizontally or vertically irregular). The damping systems, however, may be used to meet the drift limits (whether the structure is regular or irregular). Having the SFRS designed for a minimum of 75% of the strength required for structures without a DS provides safety in the event of damping system malfunction and produces a composite system with sufficient stiffness and strength to have controlled lateral displacement response.

The analysis and design of the SFRS under the base shear, V_{min} , from Eqs. (18.2-1) or (18.2-2) or, if the exception applies, under the unreduced base shear, V , should be based on a model of the SFRS that excludes the damping system.

C18.2.1.2 Damping System. The DS must be designed for the actual (unreduced) MCE_R forces (such as peak force occurring in damping devices) and deflections. For certain elements of the DS (such as the connections or the members into which the damping devices frame), other than damping devices, limited yielding is permitted provided that such behavior does not affect damping

system function or exceed the amount permitted for elements of conventional structures by the standard.

Furthermore, force-controlled actions in elements of the DS must consider seismic forces that are 1.2 times the computed average MCE_R response. Note that this increase is applied for each *element action*, rather than for each *element*. Force-controlled actions are associated with brittle failure modes where inelastic deformation capacity cannot be ensured. The 20% increase in seismic force for these actions is required to safeguard against undesirable behavior.

C18.2.2 Seismic Ground Motion Criteria. It is likely that many projects incorporating a supplemental damping system simply use design earthquake (DE) and MCE_R spectra based on the mapped values referenced in Chapter 11. Site-specific spectra are always permitted and must be used for structures on Site Class F.

When nonlinear response history analysis is used, ground motions are selected, scaled or matched and applied in accordance with the procedures of Chapter 16, with the exception that a minimum of 7 rather than 11 ground motions are required. The use of 7 motions is consistent with current practice for design of code-compliant structures, and 7 is considered an adequate number to estimate the mean response for a given hazard level. No other provisions of Chapter 16 apply to structures incorporating supplemental damping systems.

C18.2.3 Procedure Selection. The nonlinear response history procedure for structures incorporating supplemental damping systems is the preferred procedure, and Chapter 18 is structured accordingly. This method, consistent with the majority of current practice, provides the most realistic predictions of the seismic response of the combined SFRS and DS. If the nonlinear response history procedure is adopted, the relevant sections of Chapter 18 are 18.1 through 18.6.

However, via the exception, response spectrum (RS) and equivalent lateral force (ELF) analysis methods can be used for design of structures with damping systems that meet certain configuration and other limiting criteria (for example, at least two damping devices at each story configured to resist torsion). In such cases, additional nonlinear response history analysis is used to confirm peak responses when the structure is located at a site with S_1 greater than or equal to 0.6. The analysis methods of damped structures are based on nonlinear static “pushover” characterization of the structure and calculation of peak response using effective (secant) stiffness and effective damping properties of the first (pushover) mode in the direction of interest. These concepts are used in Chapter 17 to characterize the force-deflection properties of isolation systems, modified to incorporate explicitly the effects of ductility demand (post-yield response) and higher mode response of structures with dampers. Similar to conventional structures, damped structures generally

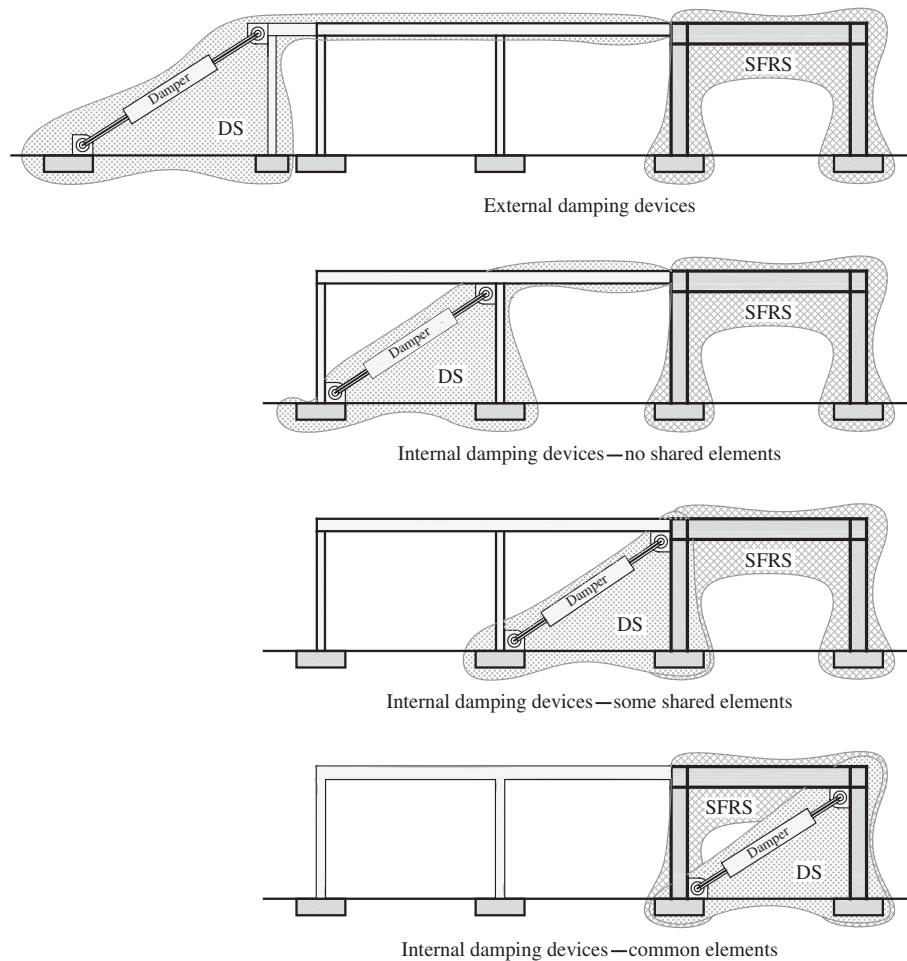


FIGURE C18.1-1 Damping System (DS) and Seismic Force-Resisting System (SFRS) Configurations

yield during strong ground shaking, and their performance can be influenced strongly by response of higher modes.

The RS and ELF procedures presented in Chapter 18 have several simplifications and limits, outlined as follows:

1. A multiple-degree-of-freedom (MDOF) structure with a damping system can be transformed into equivalent single-degree-of-freedom (SDOF) systems using modal decomposition procedures. This procedure assumes that the collapse mechanism for the structure is an SDOF mechanism so that the drift distribution over height can be estimated reasonably using either the first mode shape or another profile, such as an inverted triangle. Such procedures do not strictly apply to either yielding buildings or buildings that are nonproportionally damped.
2. The response of an inelastic SDOF system can be estimated using equivalent linear properties and a 5% damped response spectrum. Spectra for damping greater than 5% can be established using damping coefficients, and velocity-dependent forces can be established either by using the pseudovelocity and modal information or by applying correction factors to the pseudovelocity.
3. The nonlinear response of the structure can be represented by a bilinear hysteretic relationship with zero postelastic stiffness (elastoplastic behavior).
4. The yield strength of the structure can be estimated either by performing simple plastic analysis or by using the

specified minimum seismic base shear and values of R , Ω_0 , and C_d .

5. Higher modes need to be considered in the equivalent lateral force procedure to capture their effects on velocity-dependent forces. This requirement is reflected in the residual mode procedure.

FEMA 440 (2005) presents a review of simplified procedures for the analysis of yielding structures. The combined effects of the simplifications mentioned above are reported by Ramirez et al. (2001) and Pavlou and Constantinou (2004) based on studies of three-story and six-story buildings with damping systems designed by the procedures of the standard. The RS and ELF procedures of the standard are found to provide conservative predictions of drift and predictions of damper forces and member actions that are of acceptable accuracy when compared to results of nonlinear dynamic response history analysis. When designed in accordance with the standard, structures with damping systems are expected to have structural performance at least as good as that of structures without damping systems. Pavlou and Constantinou (2006) report that structures with damping systems designed in accordance with the standard provide the benefit of reduced secondary system response, although this benefit is restricted to systems with added viscous damping.

If either the RS or ELF procedures are adopted, the relevant sections of Chapter 18 are Sections 18.1, 18.2, 18.5, 18.6, and 18.7.

C18.2.4.1 Device Design. Damping devices may operate on a variety of principles and may use materials that affect their short-term and long-term performance. This commentary provides guidance on the behavior of some of these devices in order to justify the language in the standard and in order to assist the engineer in deciding on the upper and lower bound values of mechanical properties of the devices for use in analysis and design.

Damping devices that have found applications or have potential for application may be classified as follows:

1. Fluid viscous dampers (or oil dampers) that operate on the principle of orificing of fluid, typically some form of oil (Constantinou et al. 2007). These devices are typically highly engineered and precision made so that their properties are known within a narrow range. That is, when the devices are tested, their properties show small variability. One issue is heating that may have significant effects (Makris et al. 1998), which can be alleviated or eliminated by using accumulators or by using materials with varying thermal expansion properties so that the orifice size is automatically adjusted with varying temperature.

However, their long-term behavior may be affected by a variety of potential problems:

- a. Devices using accumulators include valves that may fail over time depending on the quality of construction and history of operation. It is not possible to know if and when a valve may fail.
 - b. Fluid is maintained in the device by seals between the body and the moving piston of the device, which may leak either as a result of wear caused by excessive cumulative travel or poor construction. For buildings, excessive cumulative travel is rarely an issue. When seals leak, the output of the device reduces, depending on the reduction of internal pressure of the device. It is recommended that potential leakage of oil not be considered in establishing lower bound values of property modification factors (as it is not possible to know) but rather a periodic inspection and maintenance program recommended by the manufacturer be used to detect problems and make corrections.
 - c. Orifices may be very small in diameter and therefore may result in clogging when impure oil is used or the oil is contaminated by particles of rubber used in the sealing of fluid in poorly constructed devices or by metal particles resulting from internal corrosion or because of oil cavitation when poor-quality materials are used. Typically, rubber should not be used in sealing and parts should be threaded rather than welded or connected by posttensioning. Larger diameter orifices should be preferred.
2. Viscoelastic fluid or solid devices. These devices operate on the principle of shearing of highly viscous fluids or viscoelastic solids. These viscous fluids and viscoelastic solids have a strong dependence of properties on frequency and temperature. These effects should be assessed by qualification testing. Their long-term behavior is determined by the behavior of the fluid or solid used, both of which are expected to harden with time. The engineer should ask the supplier for data on the aging of the material based on observations in real time. Information based on accelerated aging is not useful and should not be used (Constantinou et al. 2007).
 3. Metallic yielding devices. Yielding steel devices are typically manufactured of steel with yield properties that are

known within a narrow range. Nevertheless, the range of values of the yield strength can be determined with simple material tests. Also, testing some of the devices should be used to verify the information obtained in coupon testing. Aging is of least concern because corrosion may only slightly reduce the section geometrical properties. An inspection and maintenance program should eliminate the concern for aging.

4. Friction devices. Friction devices operate on the principle of preloaded sliding interfaces. There are two issues with such devices:

- a. The preload may reduce over time because of creep in sliding interface materials or the preloading arrangement, or wear in the sliding interface when there is substantial service-load related motion or after high-speed seismic motion. It is not possible to know what the preload may be within the lifetime of the structure, but the loss may be minimized when high-strength bolts are used and high-strength/low-wear materials are used for the sliding interface.
- b. The friction coefficient at the sliding interface may substantially change over time. The engineer is directed to Constantinou et al. (2007) for a presentation on the nature of friction and the short-term and long-term behavior of some sliding interfaces. In general, reliable and predictable results in the long-term friction may be obtained when the sliding interface consists of a highly polished metal (typically stainless steel) in contact with a nonmetallic softer material that is loaded to high pressure under confined conditions so that creep is completed in a short time. However, such interfaces also result in low friction (and thus are typically used in sliding isolation bearings). The engineer is referred to Chapter 17 and the related commentary for such cases. Desirable high friction (from a performance standpoint) may be obtained by use of metal to metal sliding interfaces. However, some of these interfaces are absolutely unreliable because they promote severe additional corrosion and they should never be used (British Standards Institution 1983). Other bimetallic interfaces have the tendency to form solid solutions or intermetallic compounds with one another when in contact without motion. This tendency leads to cold welding (very high adhesion or very high friction). Such materials are identified by compatibility charts (Rabinowicz 1995). The original Rabinowicz charts categorized pairs of metals as incompatible (low adhesion) to compatible and identical (high adhesion). Based on that characterization, identical metals and most bimetallic interfaces should be excluded from consideration in sliding interfaces. Excluding interfaces that include lead (too soft), molybdenum, silver, and gold (too expensive), only interfaces of tin–chromium, cadmium–aluminum, and copper–chromium are likely to have low adhesion. Of these, the tin–chromium interface has problems of additional corrosion (British Standards Institution 1983) and should not be used. Accordingly, only bimetallic interfaces of cadmium–aluminum and copper–chromium may be useful. The materials in these interfaces have similar hardness so that creep-related effects are expected to be important, leading to increased true area of contact and increased friction force over time (Constantinou et al. 2007). This increase leads to the conclusion that all bimetallic

interfaces result in significant changes in friction force over time that are not possible to predict, and therefore these types of interfaces should not be used.

5. Lead extrusion devices. These devices operate on the principle of extruding lead through an orifice. The behavior of the device is dependent on the rate of loading and temperature, and its force output reduces with increasing cycling because of heating effects. These effects can be quantified by testing so that the nominal properties and property modification factors can be established. Leakage of lead during the lifetime of the device is possible during operation and provided that the seals fail, although the effects cannot be expected to be significant. Leakage is preventable by use of proven sealing technologies and by qualification testing to verify (Skinner et al. 1993).

The registered design professional (RDP) must define the ambient temperature and the design temperature range. The ambient temperature is defined as the normal in-service temperature of the damping device. For devices installed in interior spaces, this temperature may be taken as 70°F, and the design temperature range could come from the project mechanical engineer. For devices installed exposed to exterior temperature variation, the ambient temperature may be taken as the annual average temperature at the site, and the design temperature range may be taken as the annual minimum and maximum temperatures. Since the design temperature range is implicitly tied to MCE_R analysis through λ factors for temperature, the use of maximum and minimum temperatures over the design life of the structure are considered too severe.

C18.2.4.4 Nominal Design Properties. Device manufacturers typically supply nominal design properties that are reasonably accurate based on previous prototype test programs. The nominal properties can be confirmed by project-specific prototype tests during either the design or construction phases of the project.

C18.2.4.5 Maximum and Minimum Damper Properties

Specification Tolerance on Nominal Design Properties. As part of the design process, it is important to recognize that there are variations in the production damper properties from the nominal properties. This difference is caused by manufacturing variation. Recommended values for the specification tolerance on the average properties of all devices of a given type and size are typically in the $\pm 10\%$ to $\pm 15\%$ range. For a $\pm 10\%$ specification tolerance, the corresponding λ factors would be $\lambda_{(spec,max)} = 1.1$ and $\lambda_{(spec,min)} = 0.9$. Variations for individual device properties may be greater than the tolerance on the average properties of all devices of a given type and size. It is recommended that the device manufacturer be consulted when establishing these tolerance values.

Property Variation (λ) Factors and Maximum and Minimum Damper Properties. Section 18.2.4.5 requires the devices to be analyzed and designed with consideration given to environmental conditions, including the effects of aging, creep, fatigue, and operating temperatures. The individual aging and environmental factors are multiplied together, and then the portion of the resulting λ factor (λ_{ae}) differing from unity is reduced by 0.75 based on the assumption that not all of the maximum/minimum aging and environmental values occur simultaneously.

Results of prototype tests may also indicate the need to address device behavior whereby tested properties differ from the nominal design properties because of test-related effects. Such behavior may include velocity effects, first cycle effects, and any other testing effects that cause behavior different from the nominal

design properties. This behavior is addressed through a testing λ factor (λ_{test}), which is a multiple of all the individual testing effects.

The specification (λ_{spec}), environmental (λ_{ae}) and testing (λ_{test}) factors are used to establish maximum (λ_{max}) and minimum (λ_{min}) damper properties for each device type and size for use in mathematical models of the damped structure in accordance with Eqs. (18.2-3a) and (18.2-3b). These factors are typically applied to whatever parameters govern the mathematical representation of the device.

It should be noted that more sophisticated mathematical models account for various property variation effects directly (e.g., velocity or temperature). When such models are used, the cumulative effect of the λ factors reduce (become closer to 1.0) since some of the typical behaviors contributing to λ_{max} and λ_{min} are already included explicitly in the model. Some effects, such as specification tolerance and aging, will likely always remain since they cannot be accounted for in mathematical models.

Example

Data from prototype testing, as defined in Section 18.6.1, are used to illustrate the λ factors and the maximum and minimum values to be used in analysis and design. The fluid viscous damper under consideration has the following nominal force-velocity constitutive relationship, with kips and inch units:

$$F = C \operatorname{sgn}(V)|V|^\alpha = 128 \operatorname{sgn}(v)|v|^{0.38}$$

The solid line in Fig. C18.2-1 depicts the nominal force-displacement relationship.

Prototype tests of damper corresponding to the following conditions were conducted:

- Force-velocity characteristic tests, all conducted at ambient temperature of 70°F.
 - 10 full cycles performed at various amplitudes.
- Temperature tests, three fully reversible cycles conducted at various velocities at the following temperatures:
 - 40°F
 - 70°F
 - 100°F

The data from prototype tests for each cycle (maximum and negative) are shown as data points in Fig. C18.2-1.

Also shown in the figure are the variations from nominal in the force-velocity relationships for this damper. The relationships are

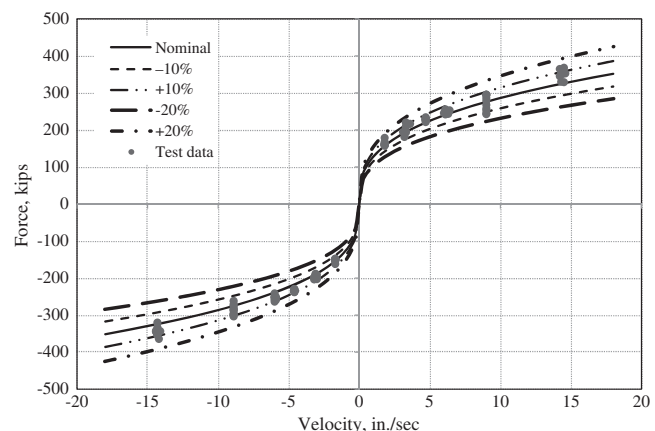


FIGURE C18.2-1 Force-Velocity Relationship for a Nonlinear Viscous Damper

obtained by changing the damper constant (C) value. No variation is considered for the velocity exponent, α . The following diagrams are shown:

- A pair of lines corresponding to damper nominal constitutive relationship computed with the C value increased or decreased by 10%. These lines account for the λ_{test} factors as defined in Section 18.2.5.4: $\lambda_{(\text{test,max})} = 1.1$, $\lambda_{(\text{test,min})} = 0.9$.

For these particular devices, the variation in properties caused by aging and environmental factors is taken as $\pm 5\%$ ($\lambda_{(ae,max)} = 1.05$, $\lambda_{(ae,min)} = 0.95$), and the specification tolerance is set at $\pm 5\%$ ($\lambda_{(\text{spec,max})} = 1.05$, $\lambda_{(\text{spec,min})} = 0.95$). These values should be developed in conjunction with the device manufacturer based on their history of production damper test data and experience with aging and other environmental effects. Using these values in Eqs. (18.2-3a) and (18.2-3b) results in $\lambda_{\text{max}} = 1.20$ and $\lambda_{\text{min}} = 0.82$. These values satisfy the minimum variation requirements of Section 18.2.4.5. They are rounded to $\lambda_{\text{max}} = 1.2$ and $\lambda_{\text{min}} = 0.8$.

- A pair of lines corresponding to the cumulative maximum and minimum λ values (accounting for testing, specification tolerance, and other factors listed in Section 18.2.4.5) computed with the nominal C value increased or decreased by 20%.

For this example, analysis with minimum and maximum damper properties should be conducted by using 80% and 120% of the nominal value for C , respectively. The analysis with maximum damper properties typically produces larger damper forces for use in the design of members and connections, whereas the analysis with minimum damper properties typically produces less total energy dissipation and hence larger drifts.

C18.2.4.6 Damping System Redundancy. This provision is intended to discourage the use of damping systems with low redundancy in any story. At least four damping devices should be provided in each principal direction, with at least two devices in each direction on each side of the center of stiffness to control torsional response. In cases where there is low damping system redundancy by this definition, all damping devices in all stories must be capable of sustaining increased displacements (with associated forces) and increased velocities (with associated displacements and forces) relative to a system with adequate redundancy. The penalty is 130%.

C18.3 NONLINEAR RESPONSE HISTORY PROCEDURE

Those elements of the SFRS and the DS that respond essentially elastically at MCE_R (based on a limit of 1.5 times the expected strength calculated using $\phi = 1$) are permitted to be modeled elastically. Modeling parameters and acceptance criteria provided in ASCE 41, with a performance objective defined in Table 2.2, as modified in this chapter, are deemed satisfactory to meet the requirements of this section.

The hardware of all damping devices (for example, the cylinder of a piston-type device) and the connections between the damping devices and the remainder of the structure must remain elastic at MCE_R (see Section 18.2.1.2). The nonlinear behavior of all other elements of both the SFRS and the DS must be modeled based on test data, which must not be extrapolated beyond the tested deformations. Strength and stiffness degradation must be included if such behavior is indicated. However, the damping system must not become nonlinear to such an extent that its function is impaired.

Table C18.3-1 Analysis Cases for Establishing Amplification Factors

Case	Damper Properties	Accidental Eccentricity
I	Minimum	No
Ila	Minimum	Yes, X direction
Ilb	Minimum	Yes, Y direction

Nonlinear response history analysis (NRHA) is performed at both the design earthquake (DE) and the MCE_R levels. Accidental eccentricity is included at MCE_R but need not be included at the DE level, since the SFRS design checks from Section 18.2.1.1 include accidental eccentricity. However, the results from the NRHA at DE using a model of the combined SFRS and DS must be used to recheck all elements of the SFRS, since the checks of Section 18.2.1.1 are conducted using a representation of the structure excluding the damping system. This requirement is defined in Section 18.4.1. The damping system is designed and evaluated based on the results of the MCE_R analyses, as defined in Section 18.4.2.

For sites classified as near-fault, individual pairs of horizontal ground motion components must be applied to the model to reflect the fault-normal and fault-parallel directions. For all other sites, each pair of horizontal ground motion components should be applied to the building at orthogonal orientations such that the mean of the component response spectra for the records applied in each direction is approximately equal (close to $\pm 10\%$) to the mean of the component response spectra of all records applied for the period range specified in Section 18.2.2.2. The design reviewer would be the judge of what constitutes “approximately equal.”

C18.3.2 Accidental Mass Eccentricity. In order to avoid the need to perform a large number of nonlinear response history analyses that include the suites of ground motions, the upper and lower bound damper properties, and five or more locations of the center of mass, the exception in this provision allows the center-of-mass analysis results to be scaled and used to account for the effects of mass eccentricity in different building quadrants.

The following is one suggested method of developing appropriate amplification factors for deformations and forces for use with center-of-mass NRHAs to account for the effects of accidental eccentricity. The use of other rationally developed amplification factors is permitted and encouraged given that the artificial shift of the center of mass changes the dynamic characteristics of the analyzed structure and may lead to the paradox of reduced torsional response with increasing accidental eccentricity (Basu et al. 2014).

The most critical directions for moving the calculated center of mass are such that the accidental eccentricity adds to the inherent eccentricity in each orthogonal direction at each level. For each of these two eccentric mass positions, and with minimum damper properties, the suite of NRHAs should be run and the results processed in accordance with Section 18.3.3. The analysis cases are defined in Table C18.3-1.

The results from Cases Ila and Ilb are then compared in turn to those from Case I. The following amplification factors (ratio of Case Ila or Ilb response to Case I response) are computed:

- The amplification for story drift in the structure at the plan location with the highest drift, enveloped over all stories;
- The amplification for frame-line shear forces at each story for the frame subjected to the maximum drift.

The larger of the two resulting scalars on drift should be used as the deformation amplifier, and the larger of the two resulting scalars on force should be used as the force amplifier. Once the amplification factors are established, the effects of accidental eccentricity should be considered as follows.

The NRHA procedure should be run for the inherent mass eccentricity case only, considering both maximum and minimum damper properties. For each damper property variation, response quantities should be computed in accordance with Section 18.3.3. All resulting deformation response quantities should be increased by the deformation amplifier, and all resulting force quantities should be increased by the force amplifier before being used for evaluation or design.

C18.4 SEISMIC LOAD CONDITIONS AND ACCEPTANCE CRITERIA FOR NONLINEAR RESPONSE HISTORY PROCEDURE

C18.4.1 Seismic Force-Resisting System. All elements of the SFRS are checked under two conditions. First, the SFRS (excluding the damping system) is checked under the minimum base shear requirements of Section 18.2.1.1. Second, the demands from the NRHA at DE (with a model of the combined SFRS and DS) must be used to recheck all elements of the SFRS.

There are three limiting values for the analytically computed drift ratios at MCE_R . Table 12.12-1 lists the allowable drifts for structures. These limiting drift ratios are checked against drift ratio demands computed from the code procedure. Since the code design is an implied DE intensity, the drift ratios in the table are also intended to be used at analysis conducted at this level.

1. 3% limit: For most common structures, the DE allowable drift ratio (Δ_d/h) is 2%. Because for most cases, the ratio of MCE_R to DE intensity is 1.5, then the allowable drift ratio at MCE_R becomes 3% ($1.5 \times 2\%$).
2. 1.9 factor: When NRHA analysis is used, the code (Section 16.2.4.3 of ASCE 7-10) allows the DE drift ratios computed from analysis to be limited to 125% of the DE drift ratio limits of Table 12.12-1. Therefore, the MCE_R drift ratios are limited to 1.9 (approximately equal to 1.5×1.25) of limits of Table 12.12-1.
3. $1.5R/C_d$ factor: The deflections δ_x of Eq. (12.8-15) are computed by amplifying the deflections computed from analysis by the deflection amplification factor (C_d). The elastic deflections used in Chapter 12 themselves are computed at DE intensity using elastic analysis with forces that are reduced by the response modification factor, R . Thus, for the purpose of comparing drift ratios computed from NRHA with Table 12.12-1, the entries of the table need to be modified by the R/C_d factor for comparison at DE level. Therefore, the allowable drift ratios at MCE_R correspond to $1.5R/C_d$ of entries of the table.

Example: Five-Story Steel Special Moment Frames in Risk Category I or II

- Allowable drift ratio from Table 12.12-1 = 2%.
- Allowable drift ratio for structures with dampers using NRHA then would be the smallest of
 - 3%,
 - $1.9 \times 2\% = 3.8\%$, and
 - $1.5 \times (8/5.5) \times 2\% = 4.4\%$.
- 3% controls. Thus, all computed drift ratios from NRHA should be 3% or less at MCE_R .

C18.5 DESIGN REVIEW

The independent design review of many structures incorporating supplemental damping may be performed adequately by one registered and appropriately experienced design professional. However, for projects involving significant or critical structures, it is recommended that a design review panel consisting of two or three registered and appropriately experienced design professionals be used.

C18.6 TESTING

C18.6.1.2 Sequence and Cycles of Testing. The use of $1/(1.5T_1)$ as the testing frequency is based on a softening of the combined SFRS and DS associated with a system ductility of approximately 2. Test 2 (d) in Section 18.6.1.2 ensures that the prototype damper is tested at the maximum force from analysis.

It should be noted that velocity-dependent devices (for example, those devices characterized by $F = C_v^a$) are not intended to be characterized as frequency-dependent under item 4 of this section.

C18.6.1.3 Testing Similar Devices. In order for existing prototype test data to be used to satisfy the requirement of Section 18.6.1, the conditions of this provision must be satisfied. It is imperative that identical manufacturing and quality control procedures be used for the preexisting prototype and the project-specific production damping devices. The precise interpretations of “similar dimensional characteristics, internal construction, and static and dynamic internal pressures” and “similar maximum strokes and forces” are left to the RDP and the design review team. However, variations in these characteristics of the preexisting prototype device beyond approximately $\pm 20\%$ from the corresponding project-specific values should be cause for concern.

C18.6.1.4 Determination of Force-Velocity-Displacement Characteristics. When determining nominal properties (item 2) for damping devices whose first-cycle test properties differ significantly from the average properties of the first three cycles, an extra cycle may be added to the test, and the nominal properties may be determined from the average value using data from the second through fourth cycles. In this case, the effect of first-cycle properties must be addressed explicitly and included in the λ_{max} factor. It should be noted that if the property variation methodology of Sections 18.2.4.4 and 18.2.4.5 is applied consistently, the maximum and minimum design properties (Eqs. (18.2-4a) and (18.2-4b)) will be identical, regardless of whether the nominal properties are taken from the average of cycles 1 through 3 or cycles 2 through 4.

C18.6.2 Production Tests. The registered design professional is responsible for defining in the project specifications the scope of the production damper test program, including the allowable variation in the average measured properties of the production damping devices. The registered design professional must decide on the acceptable variation of damper properties on a project-by-project basis. This range must agree with the specification tolerance from Section 18.2.4.5. The standard requires that all production devices of a given type and size be tested.

Individual devices may be permitted a wider variation (typically $\pm 15\%$ or $\pm 20\%$) from the nominal design properties. For example, in a device characterized by $F = C_v^a$, the mean of the force at a specified velocity for all tested devices might be permitted to vary no more than $\pm 10\%$ from the specified value of force, but the force at a specified velocity for any individual device might be permitted to vary no more than $\pm 15\%$ from the specified force.

The production dynamic cyclic test is identical (except for three versus five cycles) to one of the prototype tests of Section 18.6.1.2, so that direct comparison of production and prototype damper properties is possible.

The exception is intended to cover those devices that would undergo yielding or be otherwise damaged under the production test regime. The intent is that piston-type devices be 100% production tested, since their properties cannot be shown to meet the requirements of the project specifications without testing. For other types of damping devices, whose properties can be demonstrated to be in compliance with the project specifications by other means (for example, via material testing and a manufacturing quality control program), the dynamic cyclic testing of 100% of the devices is not required. However, in this case, the RDP must establish an alternative production test program to ensure the quality of the production devices. Such a program would typically focus on such things as manufacturing quality control procedures (identical between prototype and production devices), material testing of samples from a production run, welding procedures, and dimensional control. At least one production device must be tested at 0.67 times the MCE_R stroke at a frequency equal to $1/(1.5T_1)$, unless the complete project-specific prototype test program has been performed on an identical device. If such a test results in inelastic behavior in the device, or the device is otherwise damaged, that device cannot be used for construction.

C18.7 ALTERNATE PROCEDURES AND CORRESPONDING ACCEPTANCE CRITERIA

This section applies only to those cases where either the RS or the ELF procedure is adopted.

C18.7.1 Response-Spectrum Procedure and C18.7.2 Equivalent Lateral Force Procedure

Effective Damping. In the standard, the reduced response of a structure with a damping system is characterized by the damping coefficient, B , based on the effective damping, β , of the mode of interest. This approach is the same as that used for isolated structures. Like isolation, effective damping of the fundamental mode of a damped structure is based on the nonlinear force-deflection properties of the structure. For use with linear analysis methods, nonlinear properties of the structure are inferred from the overstrength factor, Ω_0 , and other terms.

Fig. C18.7-1 illustrates reduction in design earthquake response of the fundamental mode caused by increased effective damping (represented by coefficient, B_{1D}). The capacity curve is a plot of the nonlinear behavior of the fundamental mode in spectral acceleration-displacement coordinates. The reduction caused by damping is applied at the effective period of the fundamental mode of vibration (based on the secant stiffness).

In general, effective damping is a combination of three components:

1. Inherent Damping (β_I)—Inherent damping of the structure at or just below yield, excluding added viscous damping (typically assumed to be 2–5% of critical for structural systems without dampers).
2. Hysteretic Damping (β_H)—Postyield hysteretic damping of the seismic force-resisting system and elements of the damping system at the amplitude of interest (taken as 0% of critical at or below yield).
3. Added Viscous Damping (β_V)—The viscous component of the damping system (taken as 0% for hysteretic or friction-based damping systems).

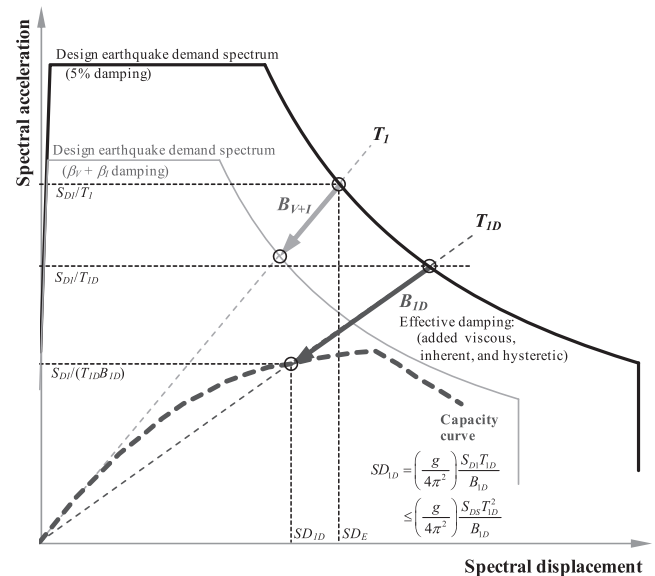


FIGURE C18.7-1 Effective Damping Reduction of Design Demand

Both hysteretic damping and added viscous damping are amplitude-dependent, and the relative contributions to total effective damping change with the amount of postyield response of the structure. For example, adding dampers to a structure decreases postyield displacement of the structure and, hence, decreases the amount of hysteretic damping provided by the seismic force-resisting system. If the displacements are reduced to the point of yield, the hysteretic component of effective damping is zero and the effective damping is equal to inherent damping plus added viscous damping. If there is no damping system (as in a conventional structure), effective damping simply equals inherent damping.

Linear Analysis Methods. The section specifies design earthquake displacements, velocities, and forces in terms of design earthquake spectral acceleration and modal properties. For equivalent lateral force (ELF) analysis, response is defined by two modes: the fundamental mode and the residual mode. The residual mode is used to approximate the combined effects of higher modes. Although typically of secondary importance to story drift, higher modes can be a significant contributor to story velocity and, hence, are important for design of velocity-dependent damping devices. For response spectrum analysis, higher modes are explicitly evaluated.

For both the ELF and the response spectrum analysis procedures, response in the fundamental mode in the direction of interest is based on assumed nonlinear (pushover) properties of the structure. Nonlinear (pushover) properties, expressed in terms of base shear and roof displacement, are related to building capacity, expressed in terms of spectral coordinates, using mass participation and other fundamental-mode factors shown in Fig. C18.7-2. The conversion concepts and factors shown in Fig. C18.7-2 are the same as those defined in Chapter 9 of ASCE/SEI 41 (2014), which addresses seismic rehabilitation of a structure with damping devices.

Where using linear analysis methods, the shape of the fundamental-mode pushover curve is not known, so an idealized elastoplastic shape is assumed, as shown in Fig. C18.7-3. The idealized pushover curve is intended to share a common point with the actual pushover curve at the design earthquake displacement, D_{1D} . The idealized curve permits definition of the global

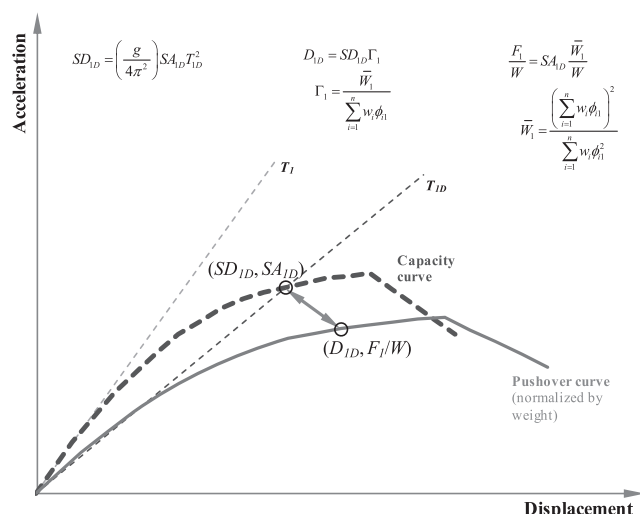


FIGURE C18.7-2 Pushover and Capacity Curves

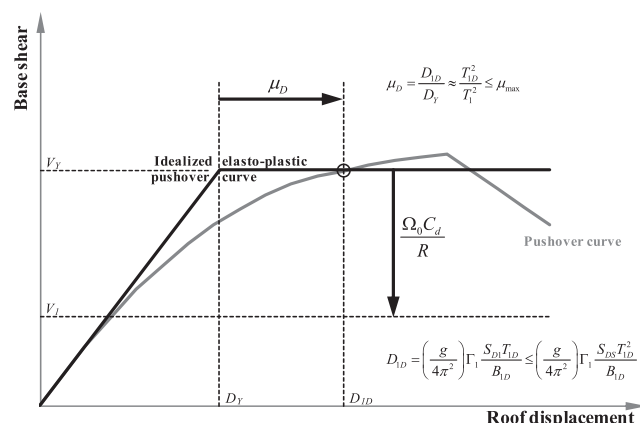


FIGURE C18.7-3 Pushover and Capacity Curves

ductility demand caused by the design earthquake, μ_D , as the ratio of design displacement, D_{1D} , to yield displacement, D_Y . This ductility factor is used to calculate various design factors; it must not exceed the ductility capacity of the seismic force-resisting system, μ_{max} , which is calculated using factors for conventional structural response. Design examples using linear analysis methods have been developed and found to compare well with the results of nonlinear time history analysis (Ramirez et al. 2001).

Elements of the damping system are designed for fundamental-mode design earthquake forces corresponding to a base shear value of V_Y (except that damping devices are designed and prototypes are tested for maximum considered earthquake response). Elements of the seismic force-resisting system are designed for reduced fundamental-mode base shear, V_1 , where force reduction is based on system overstrength (represented by Ω_0), multiplied by C_d/R for elastic analysis (where actual pushover strength is not known). Reduction using the ratio C_d/R is necessary because the standard provides values of C_d that are less than those for R . Where the two parameters have equal value and the structure is 5% damped under elastic conditions, no adjustment is necessary. Because the analysis methodology is based on calculating the actual story drifts and damping device displacements (rather than the displacements

Table C18.7-1 Values of Damping Coefficient, B

Effective Damping, β (%)	Table 17.5-1 of ASCE/SEI 7 (2010), AASHTO (2010), CBC (2013), seismically isolated structures)	Table 18.6-1 of ASCE/SEI 7 (2010) (structures with damping systems)	FEMA 440 (2005)	Eurocode 8 (2005)
2	0.8	0.8	0.8	0.8
5	1.0	1.0	1.0	1.0
10	1.2	1.2	1.2	1.2
20	1.5	1.5	1.5	1.6
30	1.7	1.8	1.8	1.9
40	1.9	2.1	2.1	2.1
50	2.0	2.4	2.4	2.3

calculated for elastic conditions at the reduced base shear and then multiplied by C_d), an adjustment is needed. Because actual story drifts are calculated, the allowable story drift limits of Table 12.12-1 are multiplied by R/C_d before use.

C18.7.3 Damped Response Modification

C18.7.3.1 Damping Coefficient. Values of the damping coefficient, B , in Table 18.7-1 for design of damped structures are the same as those in Table 17.5-1 for isolated structures at damping levels up to 20% but extend to higher damping levels based on results presented in Ramirez et al. (2001). Table C18.7-1 compares values of the damping coefficient as found in the standard and various resource documents and codes. FEMA 440 (2005) and Eurocode 8 (2005) present equations for the damping coefficient, B , whereas the other documents present values of B in tabular format.

The equation in FEMA 440 is

$$B = \frac{4}{5.6 - \ln(100\beta)}$$

The equation in Eurocode 8 (2005) is

$$B = \sqrt{\frac{0.05 + \beta}{0.10}}$$

C18.7.3.2 Effective Damping. The effective damping is calculated assuming that the structural system exhibits perfectly bilinear hysteretic behavior characterized by the effective ductility demand, μ , as described in Ramirez et al. (2001). Effective damping is adjusted using the hysteresis loop adjustment factor, q_H , which is the actual area of the hysteresis loop divided by the area of the assumed perfectly bilinear hysteretic loop. In general, values of this factor are less than unity. In Ramirez et al. (2001), expressions for this factor (which they call Quality Factor) are too complex to serve as a simple rule. Eq. (18.7-49) provides a simple estimate of this factor. The equation predicts correctly the trend in the constant acceleration domain of the response spectrum, and it is believed to be conservative for flexible structures.

C18.7.4 Seismic Load Conditions and Acceptance Criteria for RSA and ELF Procedures

C18.7.4.5 Seismic Load Conditions and Combination of Modal Responses. Seismic design forces in elements of the damping system are calculated at three distinct stages: maximum displacement, maximum velocity, and maximum acceleration.

All three stages need to be checked for structures with velocity-dependent damping systems. For displacement-dependent damping systems, the first and third stages are identical, whereas the second stage is inconsequential.

Force coefficients C_{mFD} and C_{mFV} are used to combine the effects of forces calculated at the stages of maximum displacement and maximum velocity to obtain the forces at maximum acceleration. The coefficients are presented in tabular form based on analytic expressions presented in Ramirez et al. (2001) and account for nonlinear viscous behavior and inelastic structural system behavior.

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